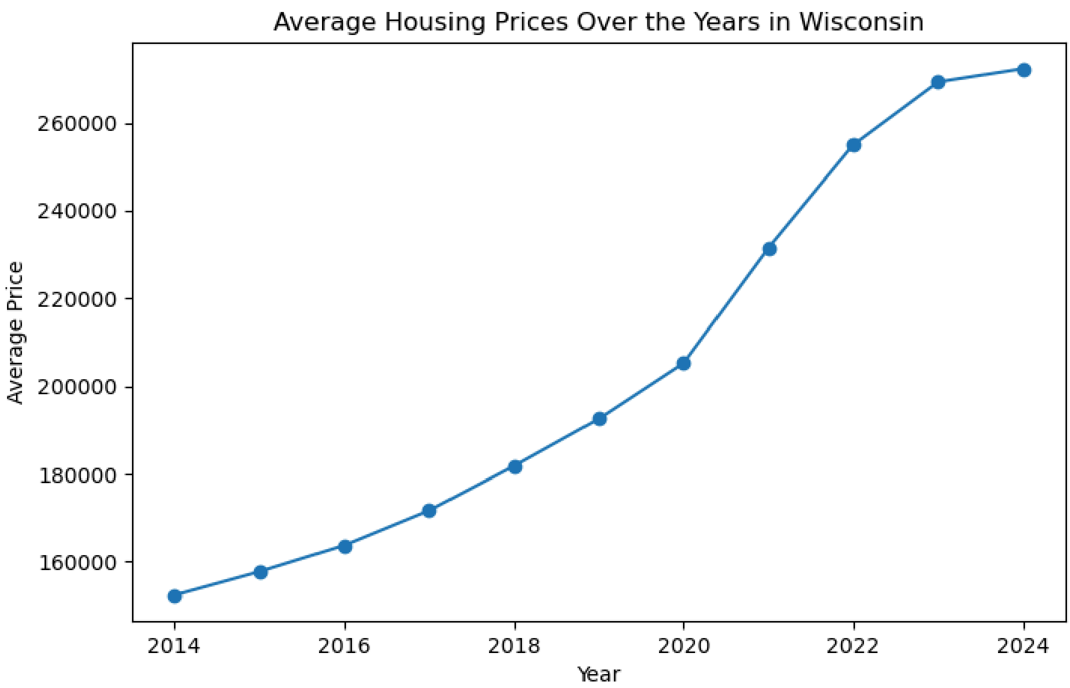
LIS705: Introduction to Analytics
Project Assignment 4

April 24, 2024

***Question 1:* What is the trend in average housing prices in Wisconsin from 2014 to 2023?**

Approach: This question involves analyzing the average housing price for each year in the dataset. The trend can be established by calculating the mean price for each year and observing the changes over time.

Data Visualization: A line chart will effectively display the trend in average housing prices across the years.

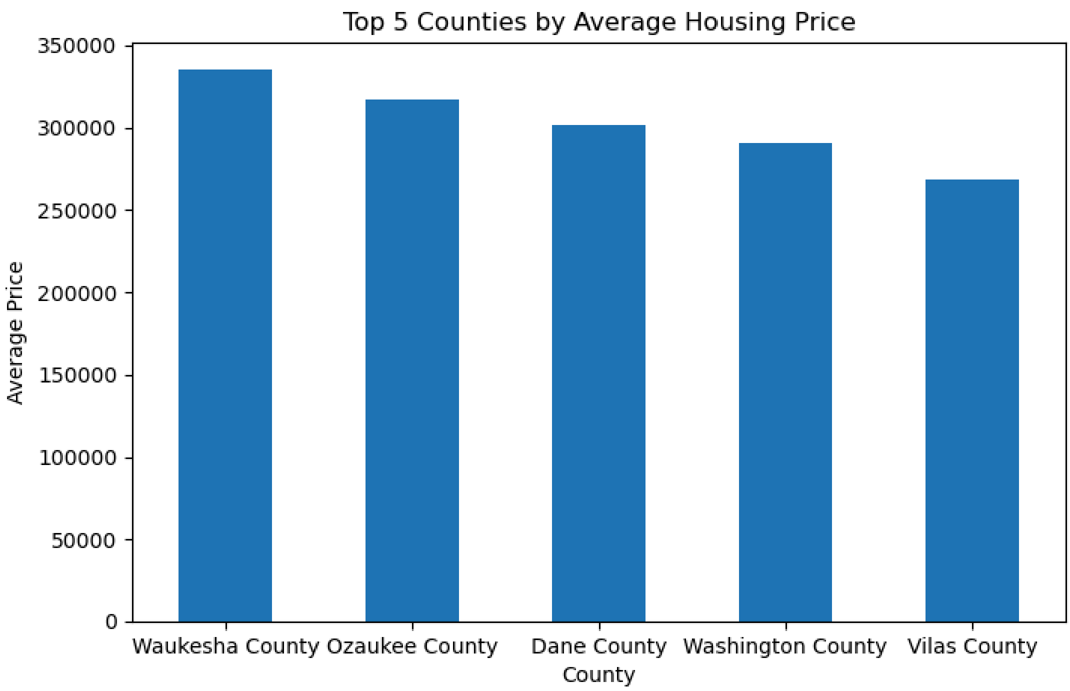


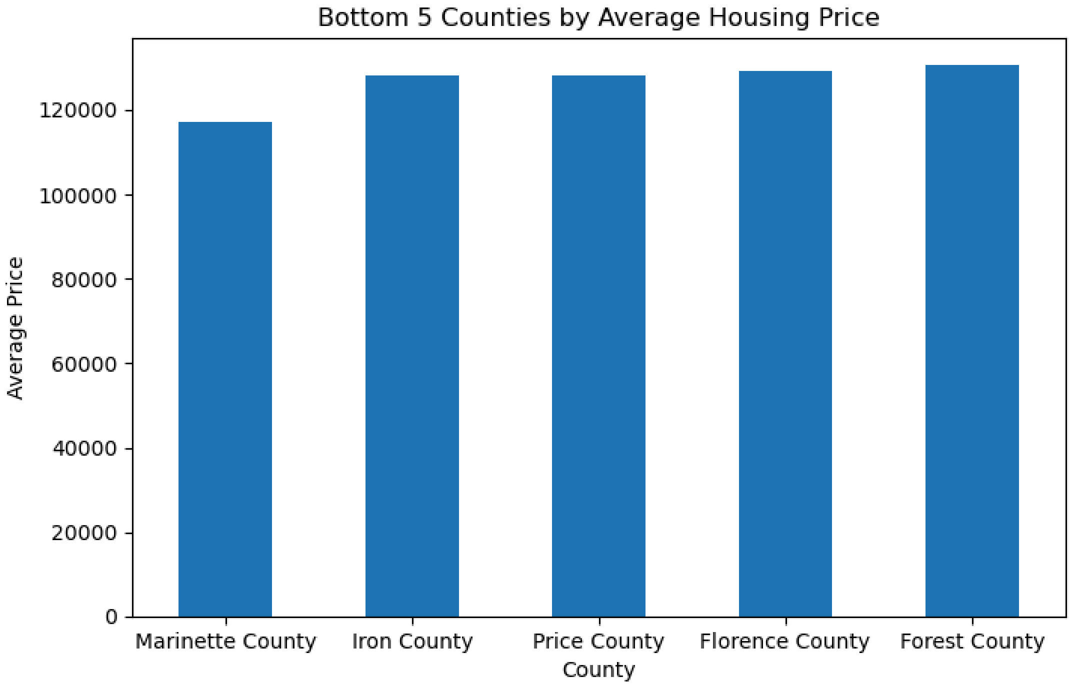
Derivative Data: You can create a new column for the year to group the data accordingly and then calculate the average price for each year.

Question 2: **Which counties in Wisconsin have the highest and lowest average housing prices from 2014 to 2023?**

Approach: This comparative analysis involves grouping the dataset by county and calculating the average price for each. It identifies the counties with the highest and lowest average prices.

Data Visualization: A bar chart can be used to compare the average housing prices across different counties, highlighting the extremes.



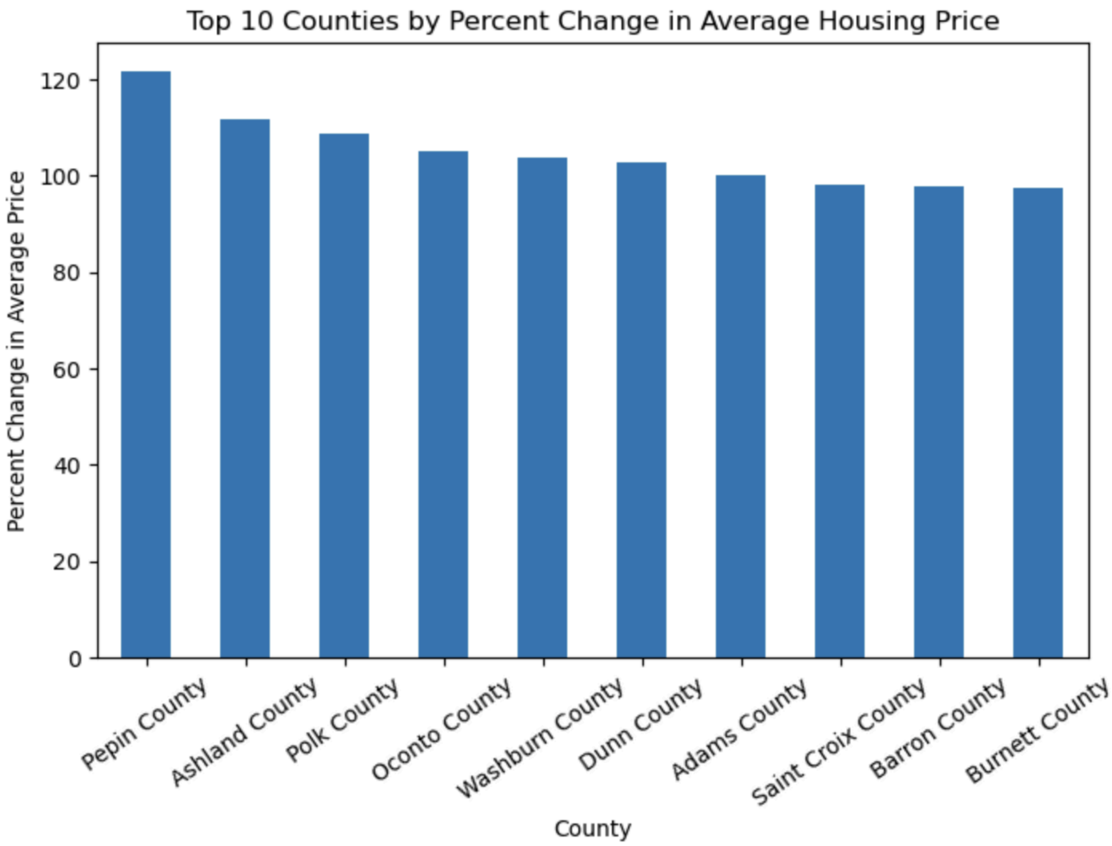


Derivative Data: You can create a new column to group the data by county, allowing you to calculate the average price for each county.

***Question 3:* What is the percent change in average housing prices from 2014 to 2023 in the top 10 counties in Wisconsin?**

Approach: Group the data by county, calculate the average housing price for each county from 2014 to 2023, and determine the top 10 counties. Calculate the percent change between 2014 and 2023 for these counties using the formula:
*percent_change = ((price_2023 - price_2014) / price_2014) * 100*

Data Visualization: Use a bar chart or line chart to visualize the percent change in housing prices for the top 10 counties. The x-axis can represent counties, while the y-axis shows the percent change.

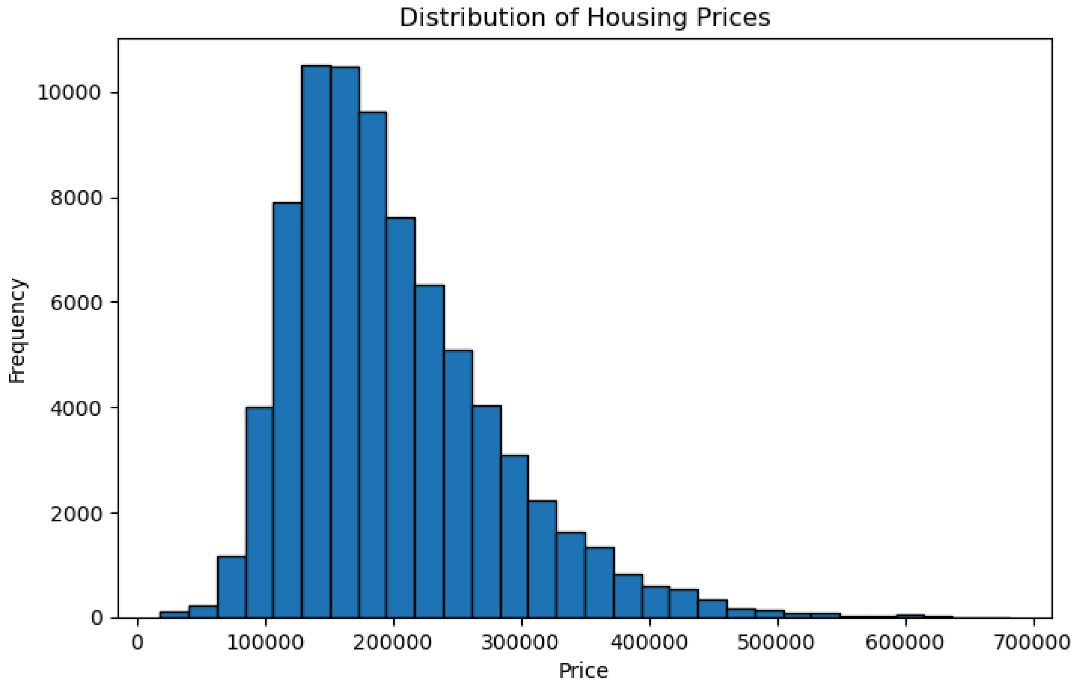


Derivative Data Fields: Create derivative fields like 'Year' to group the data, 'Top 10 Counties' to identify the counties with the highest prices, and 'Percent Change' to store the calculated values. These fields will streamline the analysis process.

Question 4: **What does the distribution of housing prices in Wisconsin from 2014 to 2023 reveal about the frequency of different price ranges, and what patterns can be observed?**

Approach: To answer this question, analyze the graph to identify common price ranges and observe the frequency of housing prices within those ranges. This involves examining the distribution to find where the majority of prices fall and identifying any outliers or unusual patterns.

Data Visualization: A histogram is an ideal visualization for showing the frequency of prices. It helps in understanding the distribution of housing prices and highlights areas of high and low frequency.



Derivative Data Fields: To derive additional insights, consider creating these derivative fields:

1. **Price Range:** Group prices into distinct ranges to identify where most prices fall.
2. **Frequency:** Represents the count of prices within each range, providing a clear picture of the distribution.